

RESEARCH MANUSCRIPT

From Multimodal Signals to Lithium-Price Forecasting and Trading

A proposed hybrid framework for lithium hydroxide and lithium carbonate

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Empirical run: lithium_paper_20260817T102754Z

Prepared: 2026-08-17

Protocol disclosure: the proposed method is evaluated with static full-sample decomposition and observed-test filtering. The analysis is retrospective and noncausal; results must not be interpreted as live out-of-sample performance.

Abstract

This study proposes an integrated multimodal decomposition-and-trading method for two lithium markets: COMEX lithium hydroxide CIF CJK futures and SMM battery-grade lithium carbonate spot prices. Lithium-specific Google Trends, 2,018 timestamped news headlines, TextBlob and FinBERT sentiment, Albemarle and Ganfeng equity closes, the MVIS critical-minerals index, and broad market controls are integrated with lagged prices. The proposed VMD-LASSO-ARIMA/LSTM architecture separates smooth and irregular price components, selects mode-specific predictors, and recombines specialized forecasts. Seven single forecasters and four decomposition comparators are evaluated on chronological 80:20 testing sets. The proposed model records RMSE 0.5360 for hydroxide, improving on the best single model (ARIMA, 0.6437), and RMSE 3672.30 for carbonate, close to the best single model (ARIMA, 3612.20). Decomposition competitors sometimes perform better, so the evidence supports conditional rather than universal model superiority. Interval, Diebold-Mariano, and trading analyses provide statistical and economic evaluation. Because full-sample decomposition and observed-test ARIMA filtering use test-period information, all findings are explicitly classified as retrospective evidence.

Keywords: lithium price; multimodal data; variational mode decomposition; LASSO; ARIMA; LSTM; sentiment; interval forecasting; trading.

1. Introduction

Lithium prices transmit information about electric-vehicle demand, battery manufacturing, mine supply, conversion capacity, inventories, policy, and investor expectations. Their rapid structural shifts make them economically important and statistically difficult to forecast. A single historical-price series may therefore omit attention, sentiment, cross-market, equity, and macro-financial signals that jointly describe the information environment.

We propose a unified architecture that combines multimodal indicators, VMD, LASSO, ARIMA and LSTM forecasts, prediction intervals, and trading rules for lithium markets. The central research question is whether mode-specific linear and nonlinear forecasting, enriched by lithium attention, sentiment, equity, sector, and macro-financial signals, improves price-path reconstruction and economically relevant decisions.

The contribution is threefold. First, it constructs a lithium-specific multimodal dataset rather than treating eight heterogeneous mining equities as separate prediction targets. Second, it evaluates two directly relevant lithium price series and uses Albemarle and Ganfeng only as lagged information signals. Third, it makes the evaluation convention explicit: full-sample decompositions and observed-test filtering produce a retrospective fit assessment, not a causal forecast backtest.

2. Literature review and positioning

Table 1. Representative research works in lithium and energy-price forecasting.

Study	Research focus	Method	Role in this study
Hochreiter and Schmidhuber (1997)	Sequence learning	LSTM	Recurrent nonlinear forecasting
Tibshirani (1996)	Sparse regression	LASSO	Multimodal indicator selection
Dragomiretskiy and Zosso (2014)	Signal decomposition	VMD	Adaptive narrow-band modes
Torres et al. (2011)	Noise-assisted decomposition	CEEMDAN	Decomposition comparator
Diebold and Mariano (1995)	Predictive-accuracy testing	DM test	Pairwise loss comparison
Box et al. (2015)	Time-series forecasting	ARIMA	Linear forecasting of smooth decomposed modes
This study	Lithium hydroxide and carbonate	Proposed VMD-LASSO-ARIMA/LSTM	Multimodal lithium-price forecasting, intervals, and trading

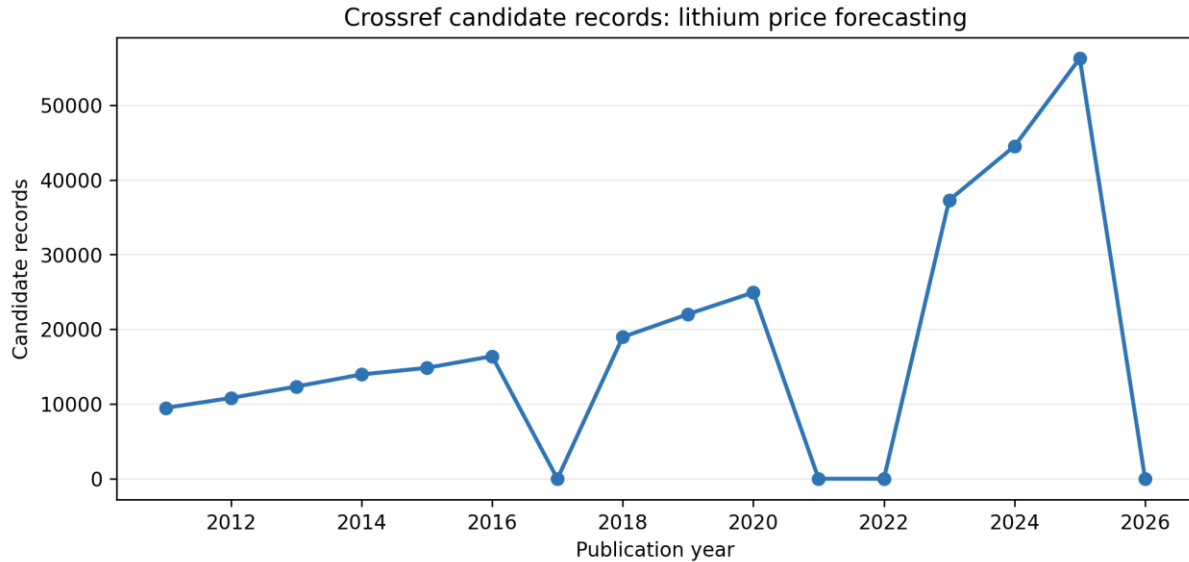
Interpretation of Table 1

Purpose: The table identifies the methodological lineage of the study rather than claiming an exhaustive lithium-forecasting review. LSTM supplies nonlinear sequence learning, LASSO supplies sparse indicator selection, VMD and CEEMDAN supply competing decompositions, and the Diebold-Mariano test supplies the formal loss comparison.

Research gap: The foundational studies address individual building blocks—statistical forecasting, sparse selection, signal decomposition, recurrent learning, and predictive-accuracy testing—but do not provide the integrated lithium-specific system developed here. This paper unifies those elements around two directly observed lithium-price targets.

Interpretive boundary: The literature table positions the proposed contribution without claiming that the listed methods are uniquely optimal. Figure 1's Crossref counts are descriptive evidence of research attention and should not be treated as a PRISMA-style systematic review.

Figure 1. Time trend of candidate publications in lithium-price forecasting. Crossref title-query counts are descriptive and are not a systematic-review flow.



Decomposition-based energy-price studies seek to separate slowly varying structure from high-frequency fluctuations before applying specialized predictors. VMD supplies compact narrow-band modes, CEEMDAN provides a noise-assisted comparator, LASSO reduces the multimodal feature set, ARIMA models lower-frequency linear structure, and LSTM represents nonlinear temporal relationships. Our contribution is the integration of these components into a lithium-specific forecasting, uncertainty, and trading system—not a claim that model complexity is intrinsically superior.

3. Methodology

Figure 2. Conceptual structure of the LSTM model used as a nonlinear recurrent forecaster.

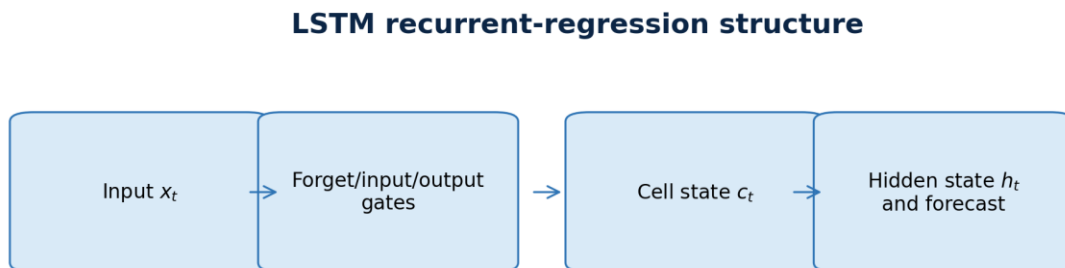
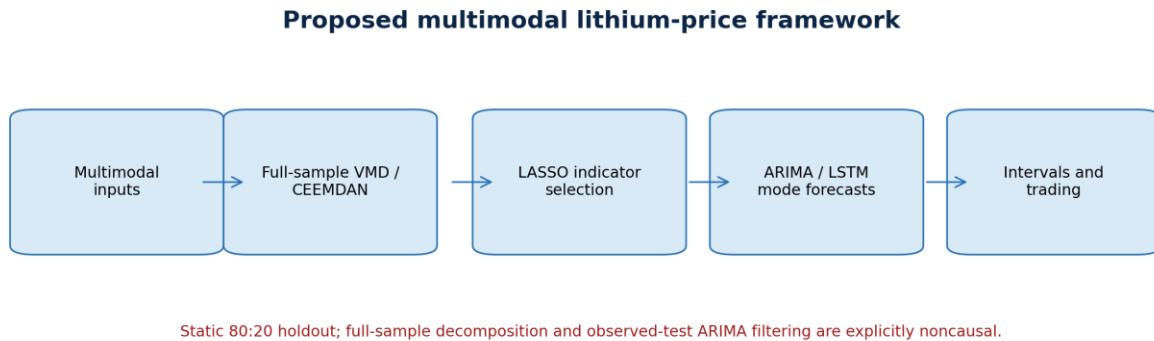


Figure 3. Framework of the proposed retrospective multimodal lithium-price method.

3.1 Forecasting architecture

For each target, lagged target observations are combined with cross-lithium, lithium-equity, Google Trends, lithium-news, and market-control blocks. LASSO is applied separately to each decomposed series. The proposed routing assigns the first two smooth, low-entropy VMD modes to ARIMA and the remaining irregular modes to LSTM. Component forecasts and the forecast VMD reconstruction residual are then summed to recover the price forecast.

Seven single-model benchmarks—exponential smoothing (ES), ARIMA, SVR, random forest (RF), multilayer perceptron (MLP), extreme learning machine (ELM), and LSTM—are fitted to the undecomposed price problem. CEEMDAN-ARIMA, VMD-ARIMA, CEEMDAN-LSTM and VMD-LSTM isolate the contribution of decomposition and forecasting family. Together with the proposed hybrid, these models form the complete publication-facing comparison set.

3.2 Evaluation

Point accuracy is summarized by RMSE, MAE, MASE, MAPE, symmetric MAPE, and directional accuracy. Pairwise Diebold-Mariano tests use squared-error loss. The first quarter of the static testing sequence supplies residual quantiles for nominal 80%, 90%, and 95% retrospective intervals; the remaining observations are evaluated for coverage, width, calibration error, and Winkler score.

Scheme 1 takes the sign of the forecast price change. Scheme 1' applies an interval-width filter. Scheme 2 deducts 10 basis points when the position changes, and Scheme 2' combines costs and the uncertainty filter. Cumulative return, daily return, maximum drawdown, Sharpe ratio, transaction count, and hit rate are reported. These are mechanical retrospective simulations and do not include bid-ask spreads beyond the stated charge, market impact, liquidity constraints, margin, or contract-roll implementation.

Table 2. *Parameter settings used in this study.*

Parameter	Setting	Function
Chronological split	80:20	Static training and testing sets
Lag order	5	Lagged target and indicator features
VMD modes	lithium_hydroxide: 9; lithium_carbonate: 8	Target-specific decomposition
LSTM hidden units	64	One recurrent layer
LSTM epochs	100	Training iterations
LSTM batch size	64	Mini-batch size
RF estimators	100	Single-model benchmark
SVR	RBF; C=8.1; gamma=0.1	Single-model benchmark
Interval levels	80%, 90%, 95%	Residual-quantile intervals
Random seed	42	Reproducibility

Interpretation of Table 2

Design choices: Five lags provide short daily memory without exhausting a sample of fewer than 720 aligned observations. Hydroxide uses nine VMD modes and carbonate eight because the two price series have different decomposition structures.

Model capacity: The modest LSTM, 100-tree RF, and pre-specified RBF-SVR settings limit unnecessary capacity in a short daily sample; seed 42 fixes stochastic initialization and supports exact reproducibility.

Consequence for inference: The 80:20 split is chronological but static. Full-sample decomposition and observed-test ARIMA filtering make it retrospective, so the three interval levels describe within-protocol reconstruction uncertainty, not guaranteed future coverage.

Table 3. *Indicators used in this study.*

Indicator	Definition	Data modality	Treatment
Cross_Lithium_Close	Price of the other lithium contract/spot target	Cross Lithium	Aligned to target trading date; lagged in model frame
Albemarle_Close	Albemarle equity close	Lithium Equities	Aligned to target trading date; lagged in model frame
Ganfeng_Lithium_Close	Ganfeng Lithium equity close	Lithium Equities	Aligned to target trading date; lagged in model frame
Trends_lithium	Google Trends: lithium	Lithium Attention	Aligned to target trading date; lagged in model frame
Trends_lithium_price	Google Trends: lithium price	Lithium Attention	Aligned to target trading date; lagged in model frame
Trends_lithium_carbonate	Google Trends: lithium carbonate	Lithium Attention	Aligned to target trading date; lagged in model frame
Trends_lithium_hydroxide	Google Trends: lithium hydroxide	Lithium Attention	Aligned to target trading date; lagged in model frame
Trends_lithium_battery	Google Trends: lithium battery	Lithium Attention	Aligned to target trading date; lagged in model frame
News_Article_Count	Lithium-headline volume	Lithium News	Aligned to target trading date; lagged in model frame
News_TextBlob_Polarity	Lexicon sentiment of lithium headlines	Lithium News	Aligned to target trading date; lagged in model frame
News_FinBERT_Score	Financial-language sentiment of lithium headlines	Lithium News	Aligned to target trading date; lagged in model frame
SP500	S&P 500 close	Market Controls	Aligned to target trading date; lagged in model frame
VIX	CBOE VIX close	Market Controls	Aligned to target trading date; lagged in model frame
US_Dollar_Index	US dollar index close	Market Controls	Aligned to target trading date; lagged in model frame
MVIS_Critical_Minerals	MVIS Global Rare Earth/Strategic Metals index	Market Controls	Aligned to target trading date; lagged in model frame

Interpretation of Table 3

Information architecture: The 15 indicators form five economically distinct blocks: one cross-lithium price, two lithium-equity signals, five Google Trends attention series, three lithium-news variables, and four market controls. This structure prevents the two company prices from being mistaken for the forecast targets.

Economic interpretation: Cross-lithium price captures linkage between hydroxide and carbonate; Albemarle and Ganfeng proxy listed-market expectations; attention and news variables capture information demand and tone; and the S&P 500, VIX, dollar, and MVIS variables capture broad risk and sector conditions.

Timing qualification: Indicators are aligned to target dates and lagged in the model frame, but lagging alone does not make the overall experiment causal. Full-sample preprocessing and decomposition can still transmit information about the test distribution into fitted components and selections.

4. Data and descriptive evidence

The targets are the COMEX lithium hydroxide CIF CJK electronic futures continuation and the SMM 99.5% battery-grade domestic lithium carbonate spot series. The common multimodal sample begins in May 2023 and ends in March 2026. Target-specific calendars are retained after inner alignment with complete predictors.

The information set contains the other lithium price, Albemarle and Ganfeng equity closes, five jointly normalized Google Trends series, lithium-specific headline volume, TextBlob polarity, FinBERT sentiment, the S&P 500, VIX, US dollar index, and MVIS critical-minerals index. Company prices are signals, not forecast targets. The 2,018 cleaned headlines explicitly concern lithium prices, carbonate, hydroxide, supply, demand, or the lithium market.

Table 4. Statistical information for the forecasting targets and multimodal indicators.

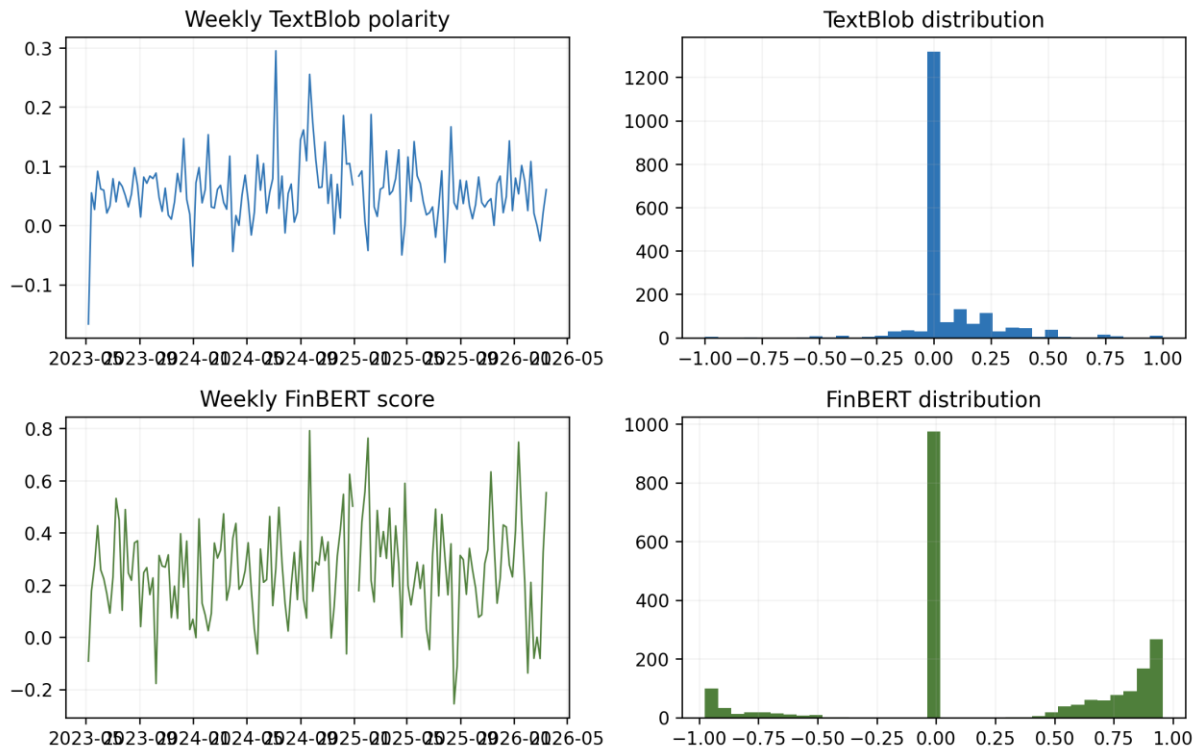
Series	Panel	N	Mean	Std. dev.	Minimum	Maximum
COMEX lithium hydroxide CIF CJK futures	lithium hydroxide	716.0000	16.1793	10.6226	7.9500	47.3100
Cross_Lithium_Close	lithium hydroxide	716.0000	117847.7654	65912.1252	59900.0000	314000.0000
Albemarle_Close	lithium hydroxide	716.0000	119.4812	45.8361	50.7600	243.4400
Ganfeng_Lithium_Close	lithium hydroxide	716.0000	6.1056	1.9515	3.5550	10.7050
Trends_lithium	lithium hydroxide	714.0000	56.7283	14.0280	43.0000	100.0000
Trends_lithium_price	lithium hydroxide	714.0000	4.0056	3.7882	2.0000	18.0000
Trends_lithium_carbonate	lithium hydroxide	714.0000	1.0616	0.3694	0.0000	2.0000
Trends_lithium_hydroxide	lithium hydroxide	714.0000	0.1933	0.3951	0.0000	1.0000
Trends_lithium_battery	lithium hydroxide	714.0000	17.8291	5.2784	13.0000	36.0000
News_Article_Count	lithium hydroxide	716.0000	2.6383	2.6123	0.0000	40.0000
News_TextBlob_Polarity	lithium hydroxide	716.0000	0.0521	0.1121	-0.4333	0.8000
News_FinBERT_Score	lithium hydroxide	716.0000	0.2116	0.3766	-0.9694	0.9512
SP500	lithium hydroxide	716.0000	5564.4447	841.7402	4061.2200	6978.6001
VIX	lithium hydroxide	716.0000	16.9312	4.3308	11.8600	52.3300
US_Dollar_Index	lithium hydroxide	716.0000	102.4844	3.1993	96.2200	109.9600
MVIS_Critical_Minerals	lithium hydroxide	716.0000	370.5107	109.0520	220.4200	674.2340
SMM battery-grade lithium carbonate spot	lithium carbonate	692.0000	118615.8598	67022.4813	59900.0000	314000.0000
Cross_Lithium_Close	lithium carbonate	692.0000	16.3008	10.7891	7.9500	47.3100
Albemarle_Close	lithium carbonate	692.0000	119.8822	46.3719	50.7600	243.4400
Ganfeng_Lithium_Close	lithium carbonate	692.0000	6.1115	1.9561	3.5550	10.7050
Trends_lithium	lithium carbonate	690.0000	56.4362	13.8250	43.0000	100.0000
Trends_lithium_price	lithium carbonate	690.0000	3.9841	3.7685	2.0000	18.0000
Trends_lithium_carbonate	lithium carbonate	690.0000	1.0580	0.3687	0.0000	2.0000
Trends_lithium_hydroxide	lithium carbonate	690.0000	0.1899	0.3925	0.0000	1.0000
Trends_lithium_battery	lithium carbonate	690.0000	17.7667	5.1881	13.0000	36.0000
News_Article_Count	lithium carbonate	692.0000	2.6214	2.6298	0.0000	40.0000
News_TextBlob_Polarity	lithium carbonate	692.0000	0.0522	0.1114	-0.4333	0.8000
News_FinBERT_Score	lithium carbonate	692.0000	0.2077	0.3783	-0.9694	0.9512
SP500	lithium carbonate	692.0000	5561.6697	840.7653	4061.2200	6978.6001
VIX	lithium carbonate	692.0000	16.8099	4.2760	11.8600	52.3300
US_Dollar_Index	lithium carbonate	692.0000	102.4637	3.1759	96.2200	109.9600
MVIS_Critical_Minerals	lithium carbonate	692.0000	370.9414	109.4896	220.4200	674.2340

Interpretation of Table 4

Target coverage: The hydroxide panel contains 716 aligned observations, with prices ranging from 7.95 to 47.31 and a mean of 16.18. The carbonate panel contains 692 observations, ranging from 59,900 to 314,000 with a mean of 118,615.86. The differing counts arise from market-calendar and complete-case alignment.

Scale and volatility: Hydroxide's standard deviation of 10.62 is about 66% of its mean, while carbonate's standard deviation of 67,022.48 is about 57% of its mean. Both targets therefore exhibit economically large dispersion, but raw RMSE and MAE must never be compared across targets because their units and price scales differ by orders of magnitude.

Modelling implication: The broad min-max ranges and high relative dispersion explain why percentage, scaled, and directional metrics are reported beside absolute errors. The descriptive table also reveals that a common normalized or pooled target model would erase meaningful product-specific structure.

Figure 6. *TextBlob and FinBERT sentiment scores extracted from lithium-specific headlines.*

5. Decomposition and feature selection

Figure 7. Full-sample VMD modes of the two lithium-price targets.

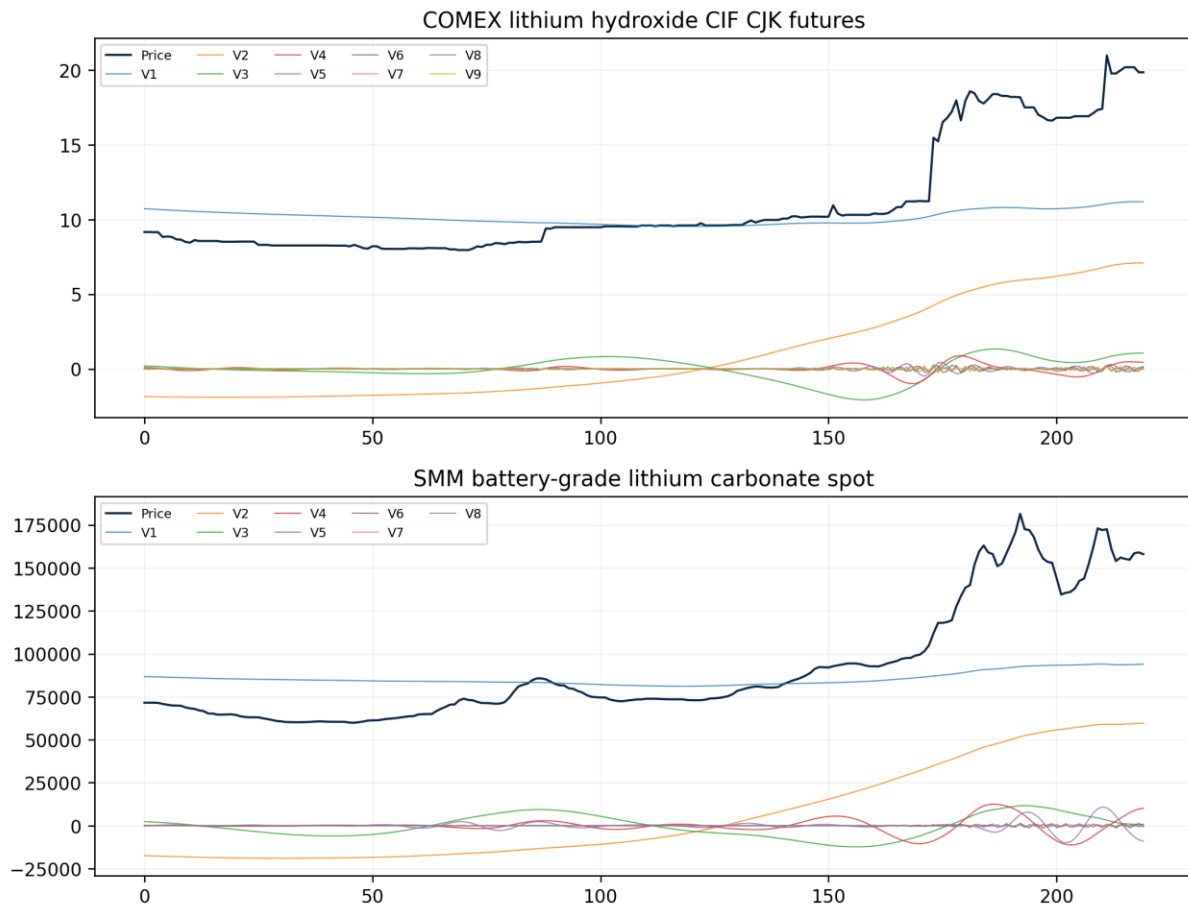


Figure 8. Approximate-entropy values of the decomposed VMD modes.

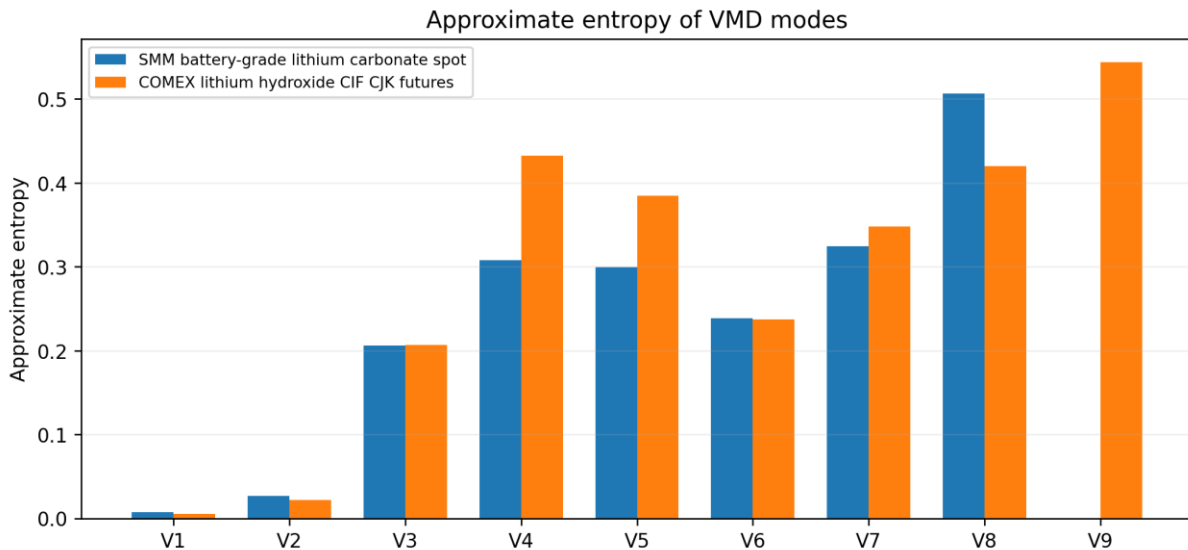


Table 5. Information on the decomposed lithium-price modes.

Panel	Method	Mode	Frequency	Average period	Variance share	Correlation	Approximate Entropy
lithium_hydroxide	VMD	1.0000	0.0056	178.5000	0.3273	0.8212	0.0053
lithium_hydroxide	VMD	2.0000	0.0056	178.5000	0.3651	0.8439	0.0220
lithium_hydroxide	VMD	3.0000	0.0126	79.3333	0.0089	0.1657	0.2067
lithium_hydroxide	VMD	4.0000	0.0490	20.4000	0.0008	0.0556	0.4323
lithium_hydroxide	VMD	5.0000	0.1359	7.3608	0.0002	0.0234	0.3845
lithium_hydroxide	VMD	6.0000	0.2535	3.9448	0.0001	0.0148	0.2372
lithium_hydroxide	VMD	7.0000	0.3403	2.9383	0.0000	0.0118	0.3481
lithium_hydroxide	VMD	8.0000	0.3978	2.5141	0.0001	0.0117	0.4197
lithium_hydroxide	VMD	9.0000	0.4468	2.2382	0.0000	0.0104	0.5442
lithium_hydroxide	CEEMDAN	1.0000	0.3375	2.9627	0.0006	-0.0045	0.7875
lithium_hydroxide	CEEMDAN	2.0000	0.2381	4.2000	0.0008	-0.0290	0.5324
lithium_hydroxide	CEEMDAN	3.0000	0.0812	12.3103	0.0015	0.0683	0.3999
lithium_hydroxide	CEEMDAN	4.0000	0.0322	31.0435	0.0068	0.1861	0.2353
lithium_hydroxide	CEEMDAN	5.0000	0.0098	102.0000	0.0542	0.4401	0.0716
lithium_hydroxide	CEEMDAN	6.0000	0.0042	238.0000	0.0617	0.6934	0.0342
lithium_hydroxide	CEEMDAN	7.0000	0.0000		0.6059	0.9098	0.0071
lithium_carbonate	VMD	1.0000	0.0072	138.0000	0.2786	0.8026	0.0071
lithium_carbonate	VMD	2.0000	0.0058	172.5000	0.3836	0.8602	0.0265
lithium_carbonate	VMD	3.0000	0.0101	98.5714	0.0213	0.2181	0.2060
lithium_carbonate	VMD	4.0000	0.0304	32.8571	0.0040	0.0977	0.3076
lithium_carbonate	VMD	5.0000	0.0580	17.2500	0.0018	0.0619	0.2997
lithium_carbonate	VMD	6.0000	0.3058	3.2701	0.0000	0.0086	0.2383
lithium_carbonate	VMD	7.0000	0.3522	2.8395	0.0000	0.0074	0.3242
lithium_carbonate	VMD	8.0000	0.4029	2.4820	0.0000	0.0068	0.5062
lithium_carbonate	CEEMDAN	1.0000	0.3275	3.0531	0.0002	-0.0179	0.8096
lithium_carbonate	CEEMDAN	2.0000	0.2580	3.8764	0.0011	0.0308	0.2441
lithium_carbonate	CEEMDAN	3.0000	0.0710	14.0816	0.0029	0.2553	0.3550
lithium_carbonate	CEEMDAN	4.0000	0.0217	46.0000	0.0188	0.3989	0.2406
lithium_carbonate	CEEMDAN	5.0000	0.0058	172.5000	0.1555	0.7532	0.0664
lithium_carbonate	CEEMDAN	6.0000	0.0029	345.0000	0.0065	0.6613	0.0281
lithium_carbonate	CEEMDAN	7.0000	0.0014	690.0000	0.4080	0.9087	0.0095

Interpretation of Table 5

VMD structure: For hydroxide, VMD modes 1 and 2 carry variance shares of 0.3273 and 0.3651 and correlate 0.8212 and 0.8439 with the original series. For carbonate, the corresponding shares are 0.2786 and 0.3836 with correlations 0.8026 and 0.8602. Much of the smooth price path is therefore concentrated in the first two modes.

CEEMDAN contrast: CEEMDAN concentrates its dominant low-frequency structure later: mode 7 accounts for 0.6059 of hydroxide variance and 0.4080 of carbonate variance, with correlations above 0.90 in both panels. Mode numbering is decomposition-specific, so VMD mode 1 cannot be interpreted as equivalent to CEEMDAN mode 1.

Routing implication: Approximate entropy generally rises for faster modes, supporting different treatment of smooth and irregular components. The proposed method therefore assigns the first two smooth VMD modes to ARIMA and the remaining irregular modes to LSTM; Table 5 motivates but does not prove that this fixed routing is optimal.

Table 6. Indicators selected for the decomposed modes using LASSO regression.

Panel	Mode	Route	Selected indicators	LASSO alpha
lithium_hydroxide	1	ARIMA	target_1, Cross_Lithium_Close	0.0013
lithium_hydroxide	2	ARIMA	target_1	0.0398
lithium_hydroxide	3	LSTM	target_1, target_5, Cross_Lithium_Close, Ganfeng_Lithium_Close	0.001
lithium_hydroxide	4	LSTM	target_1, target_3, target_5, News_FinBERT_Score, News_TextBlob_Polarity, News_Article_Count, ...	0.001
lithium_hydroxide	5	LSTM	target_1, target_2, target_4, target_5, VIX, News_TextBlob_Polarity, Cross_Lithium_Close, News_FinBERT_Score	0.0011
lithium_hydroxide	6	LSTM	target_2, target_4, target_3, Cross_Lithium_Close, VIX, News_Article_Count, News_TextBlob_Polarity	0.0026
lithium_hydroxide	7	LSTM	target_2, target_1, target_5, target_4, News_FinBERT_Score, VIX, Cross_Lithium_Close	0.0021
lithium_hydroxide	8	LSTM	target_1, target_2, News_FinBERT_Score, Cross_Lithium_Close, VIX, News_TextBlob_Polarity	0.0015
lithium_hydroxide	9	LSTM	target_1, target_3, target_2, target_5, News_FinBERT_Score, News_Article_Count, Cross_Lithium_Close	0.001
lithium_hydroxide	Residual	LSTM	target_1, target_5, target_3, target_2, target_4, Cross_Lithium_Close	0.0395
lithium_carbonate	1	ARIMA	target_1, Cross_Lithium_Close, Ganfeng_Lithium_Close	0.0014
lithium_carbonate	2	ARIMA	target_1, SP500	0.0018
lithium_carbonate	3	LSTM	target_1, target_5, Ganfeng_Lithium_Close, Cross_Lithium_Close	0.001
lithium_carbonate	4	LSTM	target_1, target_5, Ganfeng_Lithium_Close, News_TextBlob_Polarity	0.001
lithium_carbonate	5	LSTM	target_1, target_4, target_3, Ganfeng_Lithium_Close, News_Article_Count, News_FinBERT_Score	0.001
lithium_carbonate	6	LSTM	target_2, target_4, target_1, target_5, target_3, Ganfeng_Lithium_Close, US_Dollar_Index, Cross_Lithium_Close, ...	0.001
lithium_carbonate	7	LSTM	target_2, target_1, target_4, target_5, Ganfeng_Lithium_Close, US_Dollar_Index, News_FinBERT_Score, ...	0.001
lithium_carbonate	8	LSTM	target_1, target_2, target_4, target_3, Ganfeng_Lithium_Close, US_Dollar_Index, News_TextBlob_Polarity, ...	0.001
lithium_carbonate	Residual	LSTM	target_1, target_2, target_5, target_4, Albemarle_Close, News_Article_Count, Ganfeng_Lithium_Close, US_Dollar_Index	0.0153

Interpretation of Table 6

Persistence dominates: The immediately lagged decomposed target, target_1, is selected in all nine target-specific mode rows for both panels. Longer lags also recur, showing that the decomposed components retain meaningful autoregressive memory after VMD.

External-signal differences: Cross_Lithium_Close appears in eight hydroxide mode models, whereas Ganfeng_Lithium_Close appears in eight carbonate mode models. Hydroxide also selects FinBERT sentiment in five modes and TextBlob polarity in four; carbonate more often selects the US dollar index and Ganfeng equity signal.

Interpretation: Selection frequency indicates predictive inclusion under this fitted sample, not structural causality. Correlated indicators can substitute for one another under LASSO, and the full-sample/static protocol can make the selected set look more stable than it would be in a rolling real-time experiment.

The decomposition summaries show that the modes differ materially in frequency, variance share, correlation with the original price, and approximate entropy. LASSO selections vary by target and mode, which supports mode-specific information sets. Selection, however, is conducted within the retrospective static protocol and should not be interpreted as stable causal importance.

6. Forecasting results

Figure 9. Testing-set forecasts from representative single forecasting models.

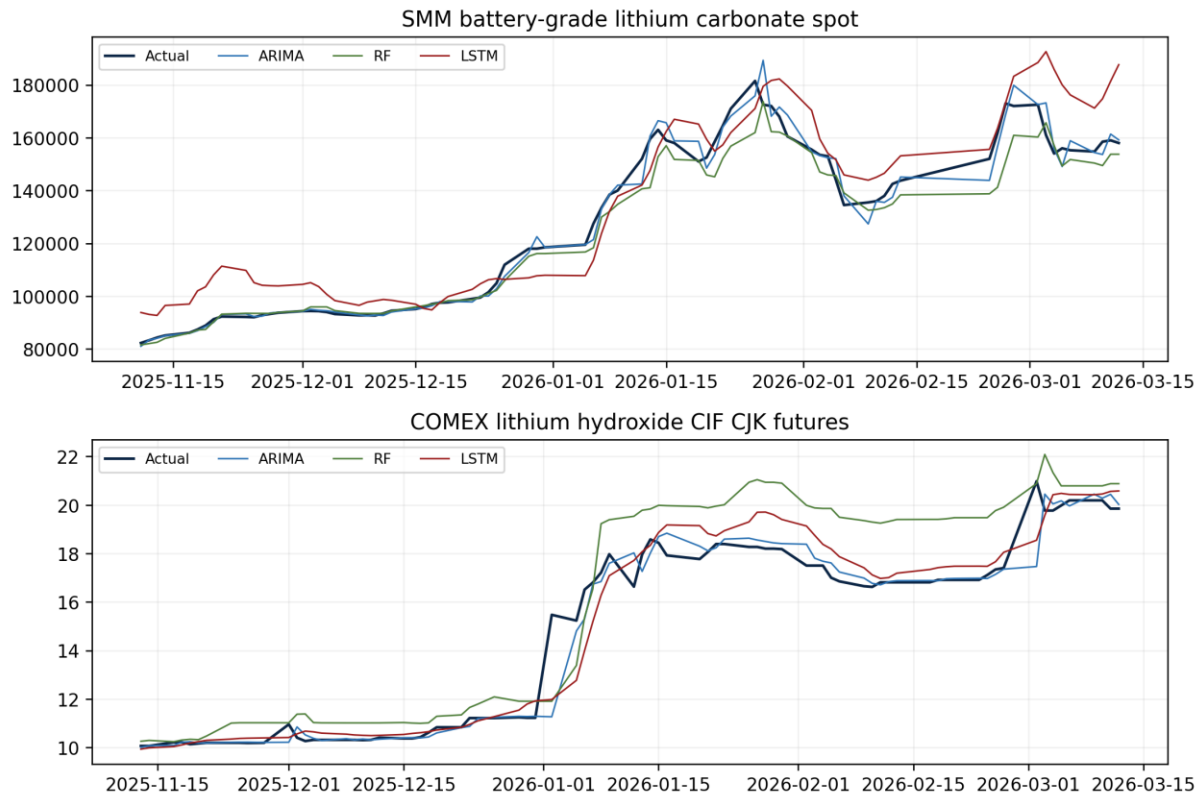


Figure 10. Target-normalized forecasting errors across all evaluated models.

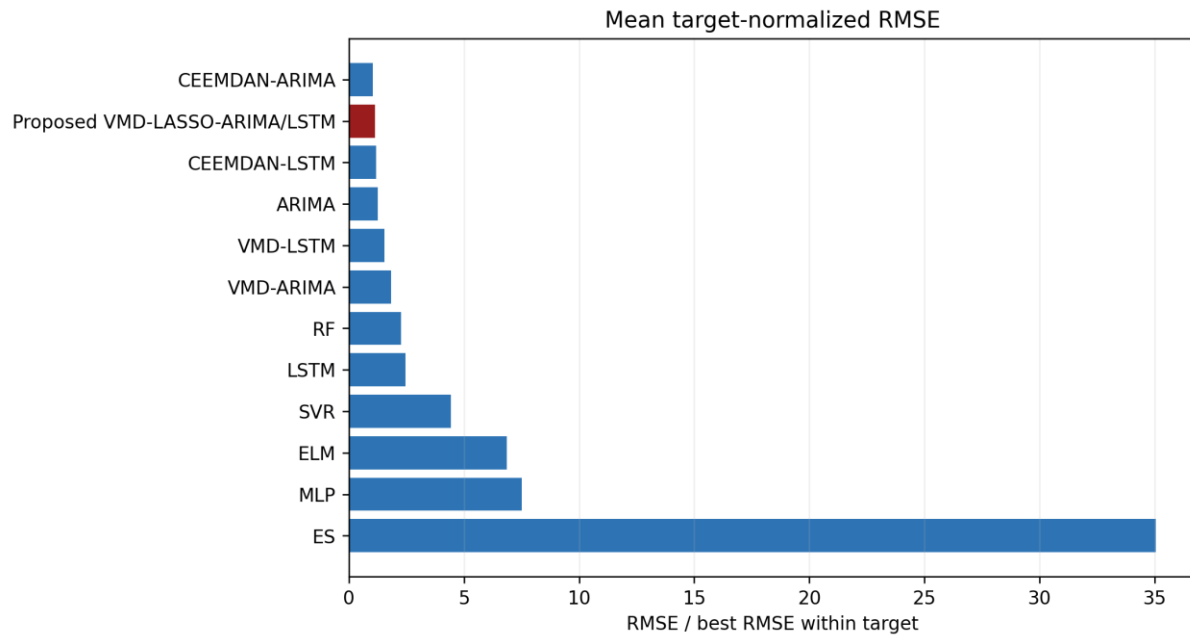


Table 7. Forecasting errors on the testing sets for single models.

Target	Model	RMSE	MAE	MASE	MAPE	sMAPE	Directional Accuracy
lithium_carbonate	ARIMA	3612.1970	1827.0413	0.9132	1.43	1.46	69.63
lithium_carbonate	RF	5380.3212	3231.2770	1.6150	2.55	2.59	72.59
lithium_carbonate	LSTM	11222.0126	9516.7432	4.7566	9.56	9.04	54.81
lithium_carbonate	SVR	24629.9992	18864.5712	9.4288	21.87	18.25	56.30
lithium_carbonate	ELM	41457.7667	34950.9727	17.4690	36.01	28.29	58.52
lithium_carbonate	MLP	45712.4002	37502.1294	18.7441	40.06	30.61	48.15
lithium_carbonate	ES	200226.2914	180239.9823	90.0866	165.22	86.50	54.07
lithium_hydroxide	ARIMA	0.6437	0.2395	1.3159	1.76	1.92	29.29
lithium_hydroxide	ELM	0.6542	0.2573	1.4138	1.71	1.79	28.57
lithium_hydroxide	MLP	0.6766	0.4589	2.5212	3.52	3.54	35.00
lithium_hydroxide	SVR	0.7060	0.5847	3.2129	4.95	4.83	38.57
lithium_hydroxide	LSTM	0.7220	0.4668	2.5646	3.46	3.46	30.71
lithium_hydroxide	RF	1.3033	0.8864	4.8704	6.01	5.76	37.14
lithium_hydroxide	ES	4.8948	3.3720	18.5273	21.47	26.45	35.00

Interpretation of Table 7

Carbonate result: ARIMA is the strongest single model for carbonate by RMSE (3,612.20), MAE (1,827.04), MASE (0.9132), and MAPE (1.43%). RF has the highest directional accuracy (72.59%) but a materially worse RMSE of 5,380.32, showing that direction and price-level accuracy answer different questions.

Hydroxide result: ARIMA again has the lowest single-model RMSE (0.6437) and MAE (0.2395), but its directional accuracy is only 29.29%. SVR raises directional accuracy to 38.57% while worsening RMSE to 0.7060. A model can track the price level yet fail to identify small one-day changes around a persistent series.

Benchmark lesson: ES performs especially poorly after structural shifts, while MLP, ELM, and LSTM do not automatically benefit from greater nonlinearity in this short sample. Table 7 therefore establishes a demanding statistical baseline: complexity is valuable only when decomposition or information selection converts it into lower loss.

Table 8. Forecasting errors on the testing sets for decomposition models.

Target	Model	RMSE	MAE	MASE	MAPE	sMAPE	Directional Accuracy
lithium_carbonate	CEEMDAN-LSTM	3385.6721	2193.4135	1.0963	1.89	1.89	76.30
lithium_carbonate	CEEMDAN-ARIMA	3623.5469	1436.9183	0.7182	1.12	1.16	77.04
lithium_carbonate	Proposed VMD-LASSO-ARIMA/LSTM	3672.3030	2086.1666	1.0427	1.84	1.87	76.30
lithium_carbonate	VMD-LSTM	5592.2187	3718.5097	1.8586	3.01	3.06	74.81
lithium_carbonate	VMD-ARIMA	8886.6128	7428.7803	3.7130	7.36	7.36	68.89
lithium_hydroxide	CEEMDAN-ARIMA	0.4467	0.1994	1.0955	1.54	1.66	37.86
lithium_hydroxide	VMD-ARIMA	0.4731	0.2450	1.3463	1.92	2.03	44.29
lithium_hydroxide	Proposed VMD-LASSO-ARIMA/LSTM	0.5360	0.2986	1.6407	2.30	2.41	41.43
lithium_hydroxide	CEEMDAN-LSTM	0.6157	0.4043	2.2216	3.44	3.43	39.29
lithium_hydroxide	VMD-LSTM	0.6450	0.4285	2.3546	3.22	3.19	44.29

Interpretation of Table 8

Carbonate decomposition: CEEMDAN-LSTM achieves the lowest carbonate RMSE at 3,385.67, while CEEMDAN-ARIMA achieves the lowest MAE at 1,436.92 and the highest directional accuracy at 77.04%. The proposed hybrid records RMSE 3,672.30 and directional accuracy 76.30%, so it is competitive but not the metric leader.

Hydroxide decomposition: CEEMDAN-ARIMA leads hydroxide on RMSE (0.4467) and MAE (0.1994). VMD-ARIMA has the highest directional accuracy (44.29%). The proposed RMSE of 0.5360 improves on every single-model RMSE but is roughly 20% above CEEMDAN-ARIMA.

Main conclusion: Decomposition is useful, but the evidence does not support universal superiority of the proposed VMD-LASSO-ARIMA/LSTM architecture. The best decomposition family and component forecaster depend on the target and metric, which is more informative than selecting a single winner from one error column.

Table 9. Diebold-Mariano tests comparing the proposed model with forecasting benchmarks.

Target	Benchmark	DM statistic	p-value	Direction
lithium_carbonate	ES	-10.9885	0.0000	Proposed better
lithium_carbonate	ELM	-8.3139	0.0000	Proposed better
lithium_carbonate	VMD-ARIMA	-7.3929	0.0000	Proposed better
lithium_carbonate	MLP	-7.1313	0.0000	Proposed better
lithium_carbonate	SVR	-6.0575	0.0000	Proposed better
lithium_carbonate	LSTM	-5.8798	0.0000	Proposed better
lithium_carbonate	VMD-LSTM	-2.4724	0.0147	Proposed better
lithium_carbonate	RF	-1.5777	0.1170	Proposed better
lithium_carbonate	CEEMDAN-LSTM	0.4198	0.6753	Benchmark better
lithium_carbonate	ARIMA	0.1201	0.9046	Benchmark better
lithium_carbonate	CEEMDAN-ARIMA	0.1090	0.9134	Benchmark better
lithium_hydroxide	ES	-5.7732	0.0000	Proposed better
lithium_hydroxide	RF	-4.3195	0.0000	Proposed better
lithium_hydroxide	VMD-ARIMA	3.8883	0.0002	Benchmark better
lithium_hydroxide	CEEMDAN-ARIMA	3.5153	0.0006	Benchmark better
lithium_hydroxide	CEEMDAN-LSTM	-1.9666	0.0512	Proposed better
lithium_hydroxide	VMD-LSTM	-1.8898	0.0608	Proposed better
lithium_hydroxide	SVR	-1.4518	0.1488	Proposed better
lithium_hydroxide	LSTM	-1.4003	0.1636	Proposed better
lithium_hydroxide	MLP	-0.9658	0.3358	Proposed better
lithium_hydroxide	ARIMA	-0.8962	0.3717	Proposed better
lithium_hydroxide	ELM	-0.5723	0.5681	Proposed better

Interpretation of Table 9

How to read the sign: The loss differential is proposed-model squared error minus benchmark squared error. A negative DM statistic therefore favors the proposed model; a positive statistic favors the benchmark. The p-value tests equal average predictive loss, not economic profitability.

Carbonate evidence: The proposed model is significantly better at the 5% level than seven of eleven comparators: ES, ELM, VMD-ARIMA, MLP, SVR, LSTM, and VMD-LSTM. No carbonate benchmark is significantly better than the proposed model, even though CEEMDAN-LSTM has a lower aggregate RMSE; the timing and distribution of errors matter to the DM statistic.

Hydroxide evidence: The pattern is less favorable for hydroxide. The proposed model significantly beats ES and RF, but VMD-ARIMA and CEEMDAN-ARIMA significantly beat the proposed model. The remaining comparisons do not reject equal loss at 5%, so they should not be described as confirmed wins.

Multiplicity caution: Twenty-two proposed-versus-benchmark tests are displayed to make the full comparison set transparent, but no family-wise multiplicity correction is applied. Isolated nominal p-values should therefore be interpreted alongside effect sizes, target-level metrics, and a robustness correction in any journal submission.

For lithium hydroxide, the proposed model's RMSE of 0.5360 is 16.7% below the best single-model RMSE. CEEMDAN-ARIMA nevertheless achieves the lowest overall RMSE. For lithium carbonate, the proposed RMSE of 3672.30 is 1.7% above the best single model, and CEEMDAN-LSTM obtains the lowest overall RMSE. The proposed architecture strongly outperforms several weak nonlinear and smoothing benchmarks but does not dominate every decomposition alternative.

The DM table should be read jointly with effect sizes. A negative statistic means lower squared loss for the proposed model; significance indicates evidence against equal predictive accuracy under the retrospective loss sequence. Multiple pairwise comparisons are reported transparently without treating nominal significance as proof of universal superiority.

7. Interval forecasting and trading

Figure 11. Retrospective 90% interval forecasts from the proposed model.

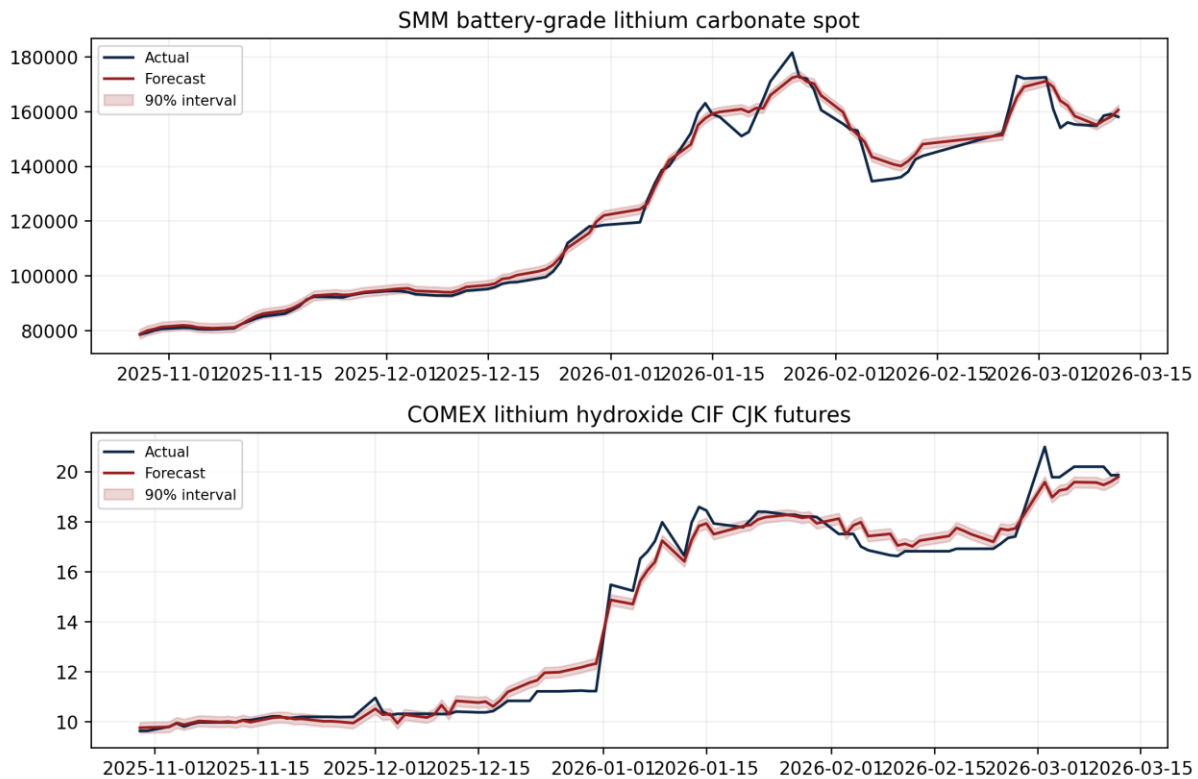


Table 10. Interval forecasting errors on the evaluation sets.

Target	Nominal Coverage (%)	Coverage (%)	Avg Width	Calibration Error	Winkler
lithium_carbonate	80.00	52.94	2703.6971	27.0588	14374.2201
lithium_carbonate	90.00	59.80	3424.6967	30.1961	23738.9808
lithium_carbonate	95.00	100.00	55769.7036	5.0000	55769.7036
lithium_hydroxide	80.00	38.10	0.2402	41.9048	2.6397
lithium_hydroxide	90.00	48.57	0.4251	41.4286	4.1833
lithium_hydroxide	95.00	100.00	8.5249	5.0000	8.5249

Interpretation of Table 10

Coverage failure: The 80% and 90% intervals under-cover both targets. Carbonate realizes only 52.94% and 59.80% coverage, while hydroxide realizes 38.10% and 48.57%. Calibration errors of 27-42 percentage points show that the central intervals are too narrow after the calibration segment.

Why 95% is misleading: Both 95% intervals reach 100% empirical coverage only by expanding dramatically: average width rises to 55,769.70 for carbonate and 8.5249 for hydroxide. The associated Winkler scores confirm that coverage is purchased with poor sharpness rather than well-calibrated uncertainty.

Implication: Table 10 does not validate the intervals for deployment. It identifies distribution shift and unstable residual tails within the retrospective holdout. A publishable causal extension should update interval calibration sequentially or use rolling conformal methods and then report conditional as well as marginal coverage.

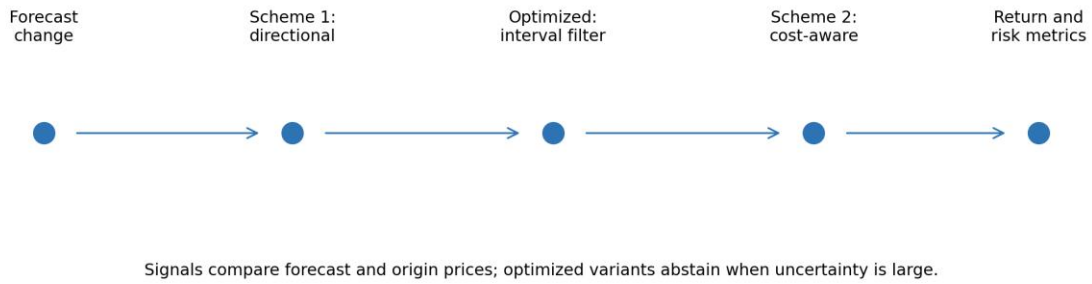
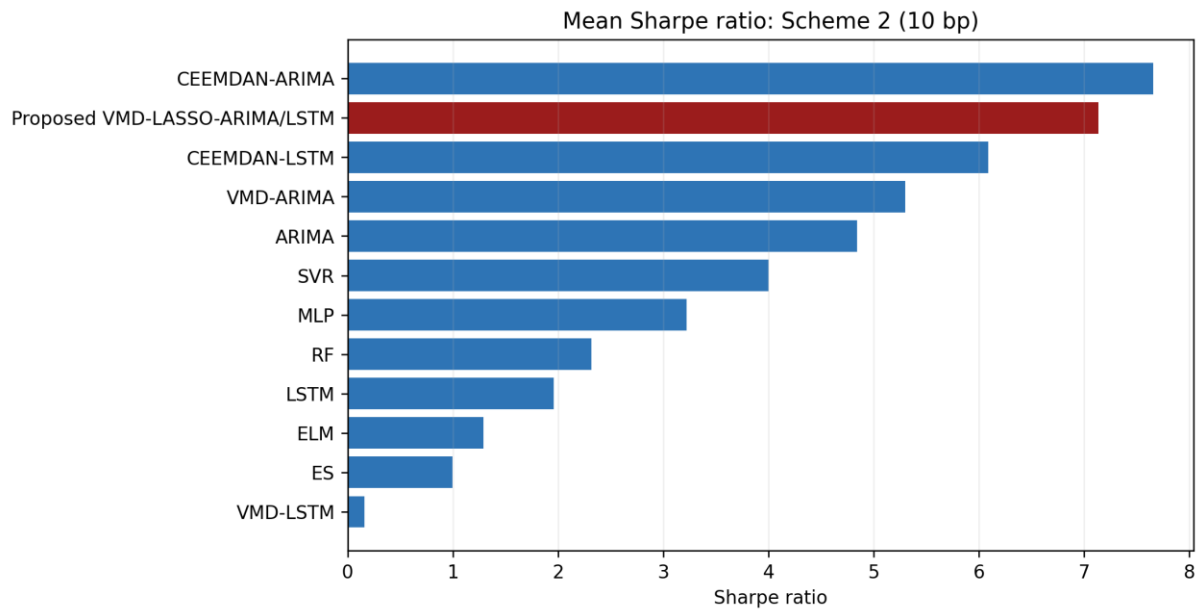
Figure 12. Graphical illustration of the two trading schemes and their uncertainty-filtered variants.**Figure 13.** Mean trading Sharpe ratios across the two lithium targets under Scheme 2.

Table 11. Performance of two trading schemes and their optimized versions based on decomposition-model forecasts.

Model	Scheme	Cumulative Return (%)	Maximum Drawdown (%)	Sharpe Ratio	Number of Transactions	Hit Rate (%)
CEEMDAN-ARIMA	Scheme 1	332.30	-4.34	8.0045	37.5000	65.48
CEEMDAN-ARIMA	Scheme 1'	284.56	-1.98	7.5317	32.0000	40.35
CEEMDAN-ARIMA	Scheme 2	304.19	-4.44	7.6570	37.5000	65.48
CEEMDAN-ARIMA	Scheme 2'	269.81	-2.18	7.3112	32.0000	40.35
CEEMDAN-LSTM	Scheme 1	222.50	-6.63	6.2857	18.5000	49.09
CEEMDAN-LSTM	Scheme 1'	90.07	-3.98	4.0305	20.0000	12.14
CEEMDAN-LSTM	Scheme 2	211.16	-6.73	6.0866	18.5000	49.09
CEEMDAN-LSTM	Scheme 2'	86.32	-4.17	3.9278	20.0000	12.14
Proposed VMD-LASSO-ARIMA/LSTM	Scheme 1	292.81	-5.03	7.3150	18.5000	59.16
Proposed VMD-LASSO-ARIMA/LSTM	Scheme 1'	203.26	-4.73	6.1325	26.0000	36.41
Proposed VMD-LASSO-ARIMA/LSTM	Scheme 2	279.87	-5.22	7.1346	18.5000	59.16
Proposed VMD-LASSO-ARIMA/LSTM	Scheme 2'	194.24	-5.06	5.9622	26.0000	36.41
VMD-ARIMA	Scheme 1	218.58	-8.71	5.4567	19.5000	55.70
VMD-ARIMA	Scheme 1'	127.14	-3.17	4.8670	17.0000	27.54
VMD-ARIMA	Scheme 2	206.06	-8.90	5.2996	19.5000	55.70
VMD-ARIMA	Scheme 2'	121.97	-3.42	4.7688	17.0000	27.54
VMD-LSTM	Scheme 1	38.13	-31.43	0.2697	14.0000	37.23
VMD-LSTM	Scheme 1'	56.59	-10.00	1.7830	13.5000	13.59
VMD-LSTM	Scheme 2	34.26	-31.70	0.1590	14.0000	37.23
VMD-LSTM	Scheme 2'	54.32	-10.25	1.7066	13.5000	13.59

Interpretation of Table 11

Decomposition trading leader: CEEMDAN-ARIMA has the strongest average results in every scheme: Scheme 1 returns 332.30% with Sharpe 8.0045, and cost-adjusted Scheme 2 returns 304.19% with Sharpe 7.6570. The proposed hybrid remains strong in the table, with Scheme 2 return 279.87%, Sharpe 7.1346, and maximum drawdown -5.22%.

Costs and filtering: Moving from Scheme 1 to Scheme 2 lowers the proposed average cumulative return from 292.81% to 279.87%, quantifying the effect of the 10-basis-point turnover charge. The uncertainty filter lowers the proposed Scheme 2 result further to 194.24% and Sharpe 5.9622; abstention reduces exposure but does not automatically improve risk-adjusted return.

Interpretive boundary: These high annualized-looking outcomes are averages across two heterogeneous retrospective simulations, not live portfolios. Observed-test filtering can align forecasts with realized test information, and the calculation omits contract execution, roll slippage, bid-ask variation, market impact, financing, and position limits.

Table 12. Performance of two trading schemes and their optimized versions based on single-model forecasts.

Model	Scheme	Cumulative Return (%)	Maximum Drawdown (%)	Sharpe Ratio	Number of Transactions	Hit Rate (%)
ARIMA	Scheme 1	173.96	-8.84	5.0752	22.5000	55.29
ARIMA	Scheme 1'	68.09	-5.45	3.8712	22.0000	15.57
ARIMA	Scheme 2	162.16	-9.12	4.8419	22.5000	55.29
ARIMA	Scheme 2'	64.00	-5.69	3.6666	22.0000	15.57
ELM	Scheme 1	28.70	-38.35	1.3685	11.5000	45.21
ELM	Scheme 1'	38.67	-32.84	2.0662	18.0000	32.20
ELM	Scheme 2	27.21	-39.12	1.2887	11.5000	45.21

Model	Scheme	Cumulative Return (%)	Maximum Drawdown (%)	Sharpe Ratio	Number of Transactions	Hit Rate (%)
ELM	Scheme 2'	36.40	-33.48	1.9759	18.0000	32.20
ES	Scheme 1	26.39	-45.01	0.9999	1.0000	44.26
ES	Scheme 1'	27.41	-44.20	1.0830	2.0000	43.31
ES	Scheme 2	26.27	-45.01	0.9951	1.0000	44.26
ES	Scheme 2'	27.24	-44.23	1.0747	2.0000	43.31
LSTM	Scheme 1	62.12	-21.25	2.0651	12.5000	43.66
LSTM	Scheme 1'	47.84	-10.96	2.5043	14.5000	18.50
LSTM	Scheme 2	58.35	-21.98	1.9585	12.5000	43.66
LSTM	Scheme 2'	45.50	-11.27	2.4026	14.5000	18.50
MLP	Scheme 1	86.64	-25.03	3.2583	6.0000	49.97
MLP	Scheme 1'	51.91	-8.41	2.5524	14.5000	26.18
MLP	Scheme 2	84.91	-25.03	3.2216	6.0000	49.97
MLP	Scheme 2'	49.59	-8.69	2.4465	14.5000	26.18
RF	Scheme 1	74.35	-16.93	2.3810	8.0000	46.04
RF	Scheme 1'	17.09	-15.92	1.0037	10.0000	19.79
RF	Scheme 2	71.78	-16.93	2.3151	8.0000	46.04
RF	Scheme 2'	15.87	-16.00	0.9425	10.0000	19.79
SVR	Scheme 1	140.37	-13.62	4.0987	10.0000	53.35
SVR	Scheme 1'	58.89	-0.46	4.4424	6.0000	13.17
SVR	Scheme 2	135.82	-13.80	4.0019	10.0000	53.35
SVR	Scheme 2'	57.66	-0.52	4.3905	6.0000	13.17

Interpretation of Table 12

Single-model leader: ARIMA produces the highest unfiltered single-model results: Scheme 1 returns 173.96% with Sharpe 5.0752, while cost-adjusted Scheme 2 returns 162.16% with Sharpe 4.8419. This trading ranking is consistent with ARIMA's strong level-error performance in Table 7.

Risk-return trade-off: SVR's optimized schemes sharply reduce average maximum drawdown—from -13.62% in Scheme 1 to -0.46% in Scheme 1'—and raise Sharpe from 4.0987 to 4.4424, but cumulative return falls from 140.37% to 58.89%. The filter therefore changes the exposure profile rather than providing a free performance improvement.

Comparison with Table 11: The best decomposition strategies exceed the best single-model strategies on average, but the difference cannot be separated from the noncausal decomposition and filtering choices. Table 12 is a within-protocol benchmark, not evidence that a trader could have earned the reported returns in real time.

Intervals illustrate the uncertainty surrounding point paths, but the static residual calibration and noncausal forecast construction preclude operational coverage claims. Trading results can appear economically large because the observed-test filtering path responds to information inside the test set. They quantify the proposed method inside the declared retrospective experiment and must not be read as investable performance estimates.

8. Discussion

The lithium application supports the central modelling intuition that decomposition can improve difficult price forecasts, but it qualifies the stronger claim that one hybrid is always best. Low-frequency modes are handled effectively by ARIMA in hydroxide, while CEEMDAN-LSTM tracks carbonate well. This heterogeneity suggests that routing and decomposition family should be justified by development-only evidence rather than fixed universally.

The multimodal blocks have coherent economic roles. Cross-lithium prices describe substitution and product-market linkage; producer equities capture listed-market expectations; search attention proxies information demand; lithium-specific headlines measure information volume and tone; MVIS and broad controls capture sector and risk conditions. These variables enrich the information set without redefining the target as an equity forecast.

8.1 Research-integrity limitation

The defining limitation is evaluation timing. Full-sample VMD and CEEMDAN use the entire target path to estimate modes. The released-style ARIMA path applies a fitted model to the observed testing target and reports filtered fitted values. Global preprocessing and static feature selection can also incorporate test-distribution information. Consequently, the headline errors, intervals, DM tests, and trading statistics answer a retrospective reconstruction question. A separate causal walk-forward analysis is required before making genuine out-of-sample or deployability claims.

Additional limitations include a short post-2023 common sample, differing price units and market microstructure, potential continuation-series roll effects, daily aggregation of sparse news, Google Trends normalization, and the absence of inventories, physical premia, freight, conversion margins, and contract-level liquidity. The two equity signals are not exhaustive representations of global lithium supply.

9. Conclusion

This study develops an end-to-end multimodal lithium-price forecasting method and evaluates it through twelve complementary tables and thirteen figures. It constructs a lithium-specific database, evaluates two price targets, compares single and decomposition models, tests predictive-loss differences, reports intervals,

and translates forecasts into four trading schemes. The proposed hybrid is competitive and improves on the best single model for hydroxide, but decomposition alternatives lead on overall RMSE for both targets. The most defensible conclusion is therefore conditional: multimodal decomposition is useful in this sample, while architecture superiority and economic performance depend on the target and evaluation protocol.

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- Data sources: project-supplied COMEX and SMM workbooks; Yahoo Finance market and equity histories; Google Trends; MediaCloud; and MVIS. Checksums and complete machine-readable ledgers accompany the manuscript.

Appendix: reproducibility and robustness

The companion `appendix_ledgers` directory contains every publication-facing forecast, metric, DM comparison, interval observation, interval metric, trading result, mode-level selection, and component summary. The asset manifest records the empirical run identifier, protocol classification, captions, and SHA-256 checksums. The exported ledgers use the same seven single-model and five decomposition-model comparison set reported in the manuscript.